POLAR HERB ATLAS

Search herb, function, compound or keyword...

Developer Dashboard

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PolarHerb® ATLAS #0001

Cordyceps Sinensis

(Ophiocordyceps sinensis)

Altitude Range: 3,800 - 5,500m | Region: Tibetan Plateau (Yushu)

Functional Domains

Domain	Score
Energy	5.0
Respiratory	4.5
Immunity	4.0
Longevity	3.5
Cognitive	3.0

Development Score

85 / 100

A flagship high-altitude bioactive for energy and resilience, built for premium products where rarity justifies value.

Product Snapshots

FUNCTIONAL BEVERAGE
Performance Shot

DIETARY SUPPLEMENT
Performance Capsules

FUNCTIONAL INGREDIENT
Standardized Extract

DERMO-COSMETIC ACTIVE
Cellular Energy Complex

IDEAL FOR

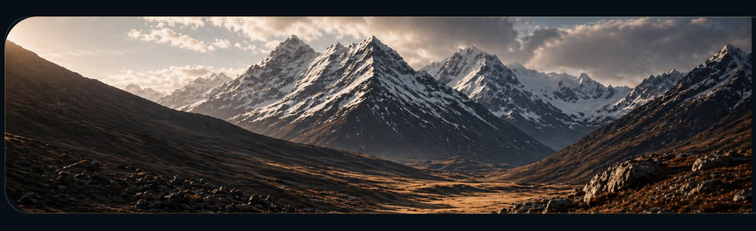
PERFORMANCE
- Sports nutrition brands
- Functional beverage lines

WELLNESS
- Supplement & Longevity products

COSMETIC
- Anti-fatigue skincare

Origin Narratives

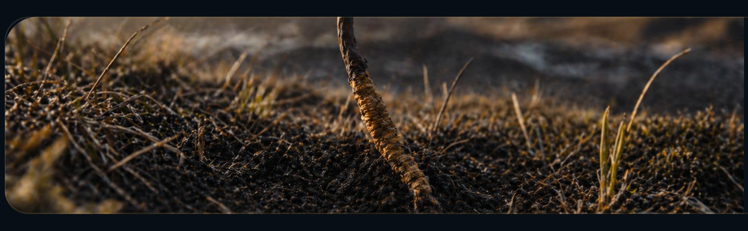
The story behind Cordyceps sinensis — shaped by altitude, rarity, and tradition.



EXTREME ENVIRONMENT

Life at the Edge of Survivial

The Tibetan Plateau rises above 4,000 meters, where oxygen thins, temperatures drop, and life slows to its limits.
Cordyceps sinensis emerges in this extreme environment - not despite the conditions, but because of them.



BIOLOGICAL RARITY

A Rare Union of Nature

Unlike conventional plants, Cordyceps is a fusion of fungus and host, developing underground for years before emerging
Its growth is slow, unpredictable, and rare - making each specimen a biological exception rather than a scalable crop.

CULTURAL VALUE

A Heritage of Vitality

For centuries, Cordyceps has been regarded as one of the most valuable natural substances across the Himalayan region.
Its rarity, combined with perceived vitality benefits, has positioned it as both a cultural symbol and a high-value natural resource.

Bioactive Intelligence

From molecule to function — AI-predicted pathways of Cordyceps sinensis.

APPLICATION CONTEXT

Functional Nutrition | Performance | Skincare/Cosmetic | Sensory

KEY BIOACTIVES

Cordycepin | Adenosine | D-Mannitol | Ergosterol

AI integrates multi-omics, pharmacokinetic and systems biology to predict how key compounds work in the human body.

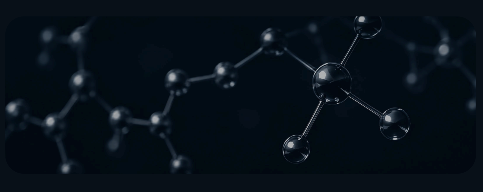
STAGE 01

Absorption & Survivability

AI-predicted pharmacokinetic analysis suggests that several Cordyceps compounds may remain biologically active after oral intake, supporting circulation and systemic delivery.

STAGE 02

Target Interaction



AI-predicted target interaction suggests that Cordyceps compounds may influence pathways linked to oxygen efficiency, cellular energy metabolism and fatigue adaptation.

STAGE 03

Functional Outcome

AI pathway convergence indicates potential support for endurance, oxygen utilization and resilience under high-stress physiological conditions.

BIOACTIVE INSIGHT

Cordyceps appears biologically aligned with high-altitude stress adaptation, particularly in pathways related to oxygen efficiency and endurance resilience.

Adapted to Hypoxia | Energy Endurance | Resilience Support

AI-PREDICTED USING:

- ADMET Predictor — PK & toxicity modeling
- SwissTargetPrediction — Target probability scoring
- DiffDock — Molecular docking
- KEGG Pathway — Pathway enrichment
- Literature Mining — Evidence integration

Open in Developer Dashboard — Explore detailed molecular data, targets, pathways, docking results and experimental evidence in the Atlas Developer Dashboard.

Comparative Matchup

Cordyceps Sinensis	VS	Ginseng
85	Oxygen Efficiency	62
80	Energy Metabolism	78
88	Adaptogenic Strength	70
72	Bioavailability	82
95	Rarity	48

Cordyceps Sinensis — Altitude Range: 3,800 - 5,500m; Core Trait: High-altitude oxygen adaptation

Ginseng — Altitude Range: 300 - 500m; Core Trait: Broad systemic resilience support

Select challenger herb:

Ginseng | Gymnadenia conopsea | Rhodiola rosea | Potentilla anserina

MATCHUP INSIGHT

Cordyceps excels in oxygen utilization, energy metabolism, and adaptogenic strength under high-altitude conditions.
Ginseng provides broader systemic support and is more accessible in supply.

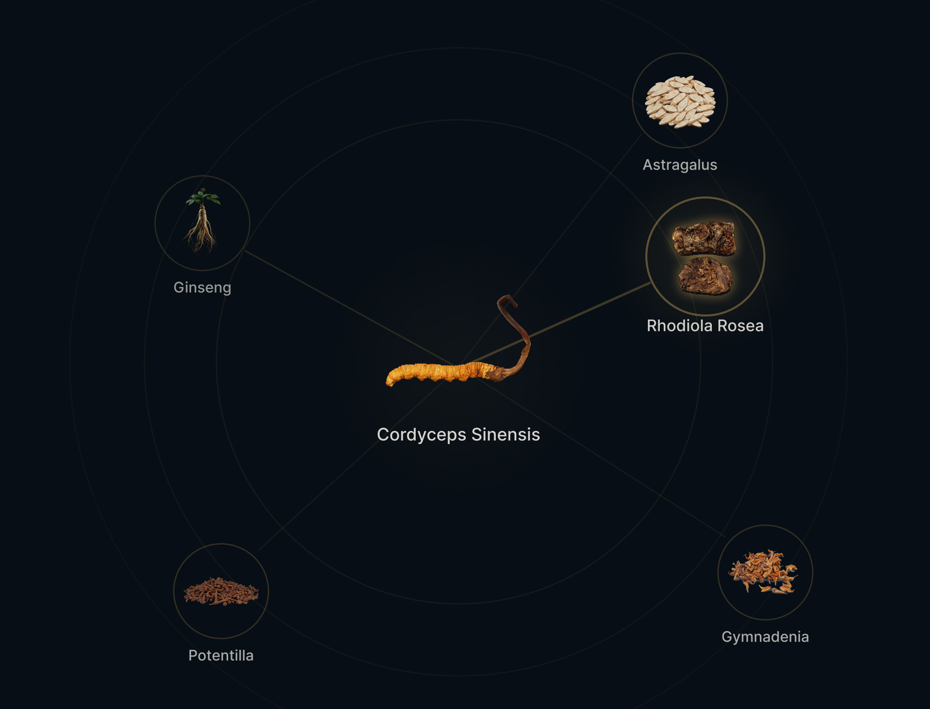
Synergy Network

Explore how Cordyceps Sinensis works in synergy with other herbs.

APPLICATION CONTEXT

Functional Nutrition | Performance | Skincare / Cosmetic | Sensory

Select a pairing herb to explore synergy.

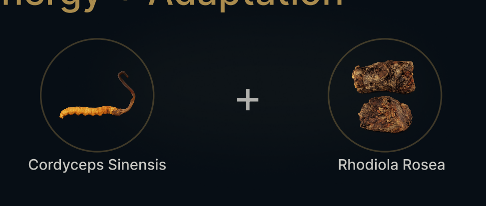


Active Stack

01 / 05

Energy | Recovery | Endurance | Adaptation | Respiratory

Energy + Adaptation



KEY SYNERGETIC EFFECTS

- **Enhances ATP Production** — Supports cellular energy generation and recovery.
- **Improves Oxygen Utilization** — Enhances oxygen uptake and efficiency under stress.
- **Reduces Fatigue & Improves Endurance** — Supports physical and mental stamina.

Evidence Level: High

Suggested Applications: Energy | Adaptation | Endurance

Formulation Readiness

KEY FORMULATION CONSIDERATIONS

Consideration	Description	Score
Formulation Compatibility	Water extracts integrate well with most beverage, capsule, and functional food systems.	4.0 / 5
Effective Dosage Practicality	Functional benefits typically achieved within 1-3 g/day of extract, fitting standard servings.	4.0 / 5
Sensory Burden	Mild earthy aroma and slight bitterness; minimal impact in most formulations.	4.0 / 5
Processing Resilience	Stable under drying, heat processing, and storage with proper handling.	4.0 / 5

SUITABLE PRODUCT FORMATS

Format	Suitability
Powder / Drink Mix	Very Suitable
Capsule / Tablet	Very Suitable
Liquid / Tonic	Very Suitable
Functional Food	Very Suitable
Topical / Cosmetic	Possible
Sensory / Experiential	Possible

READINESS INSIGHTS

OVERALL

Commercial-Ready Ingredient

Suitable across beverage, capsule, and functional food systems.

Technically Feasible | Manufacturing Ready | Market Appropriate

Regulatory Status

Understand where Cordyceps Sinensis can be legally commercialized and to what extent across global markets.

🇺🇸 United States ▼

🇺🇸 United States

Broadly Accessible

Regulatory Status

Cordyceps Sinensis is recognized as a food ingredient. It can be used in foods, functional foods, and dietary supplements.

Permitted Uses

Functional foods, beverages, capsules, tonics, powdered blends, and dietary supplement systems.

Claims Permissibility

Structure-function claims may be used within applicable dietary supplement and food regulatory frameworks.

Overall Commercialization Extent

Very Limited | Limited | Moderate | Broad | Very Broad

DEPLOYMENT NOTES
- No Novel Food Barrier
- Functional Claims Possible
- Supplement Category Compatible

Note: Regulatory status may evolve. Always confirm with up-to-date local regulations before launch.

Evidence Level

Summary of the real-world evidence supporting the PolarHerb® Atlas Profile of Cordyceps Sinensis.

EVIDENCE AT A GLANCE

- **186+ Supporting Literatures** — Peer-reviewed articles from scientific databases
- **25+ Human Studies** — Clinical trails and observational studies
- **Widely Commercialized** — Present in supplements, functional food & tonics worldwide
- **1000+ Years Traditional Use** — Long-standing use in both TCM and Folk Medicine in the Tibetan Plateau region.

EVIDENCE COMPOSITION

Area	Rating	Description
Traditional Knowledge	Very Strong	Extensive historical use in traditional medicine systems for vitality, energy and respiratory support.
Preclinical Research	Strong	Substantial preclinical evidence supporting immune modulation, mitochondrial function, and anti-inflammatory activities.
Human Research	Moderate	Emerging clinical and observational studies indicating benefits in exercise performance, fatigue reduction and quality of life.
Commercial Adoption	Strong	Widely used in supplements, functional foods and tonics globally with consistent market demand.
AI Mechanistic Confidence	Very Strong	AI analysis indicates plausible mechanisms aligned with known bioactivities, with moderate confidence.

EVIDENCE INSIGHT

Well-Supported Domains

Energy & endurance, oxygen utilization, respiratory support, immune modulation, and antioxidant activities.

Evidence Gaps

Long-term clinical validation, disease-specific claims, standardized dosing outcomes in diverse populations.

Next Break-Through

Well-designed human trials on performance, longevity, and specific health outcomes.

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